

Unit Circle and Trigonometric Graphs

Answer Key

Precalculus Trigonometry

Quick Answers

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|------------------------|---|
| 1. 14 | 7. 3, 9 |
| 2. 5 | 8. 4, 8, $\frac{\pi}{6}$ |
| 3. 9 | 9. 8, 10, 12 |
| 4. 8, $\frac{\pi}{4}$ | 10. 10 |
| 5. 20, $\frac{\pi}{6}$ | 11. 1, 5 |
| 6. 33 | 12. $10\sqrt{3} + 23$, $10\sqrt{3} + 23$ |
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Curriculum

Level: Precalculus TrigonometryHSA-CED.A.1–A.4 – *problems 5, 8, 12*HSF-TF.A.2, HSF-TF.B.5 – *problems 1, 2, 3, 4, 6, 7, 9, 10, 11**Difficulty 1–5 of 5 across 20 tagged checks.*

Common wrong answers

2. If they got 10: used the whole vertical rise from minimum to maximum instead of half of it
6. If they got 3.05: evaluated the cosine in degrees instead of radians
9. If they got 6: treated the distance from the maximum to the next minimum as a whole period
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1. Evaluate Curve G at $t = 2$.

Step 1. The input is given, so this is a substitution — and the angle must be read in radians, which is what $\pi t/4$ produces.

Step 2. $\frac{\pi(2)}{4} = \frac{\pi}{2}$, and $\sin \frac{\pi}{2} = 1$ from the unit circle.

Step 3. $h(2) = 5(1) + 9 = \boxed{14}$, which is the highest point on Curve G — the graph agrees.

2. Amplitude of Curve G.

Step 1. Amplitude is the distance from the midline to a peak, which is *half* the total rise from the lowest value to the highest — not the whole rise.

Step 2. $\frac{14 - 4}{2} = \frac{10}{2}$.

Step 3. The amplitude is $\boxed{5}$.

Common wrong answer: 10 — the full vertical distance between the minimum and the maximum, which is twice the amplitude.

3. Midline of Curve G.

Step 1. The midline sits exactly halfway between the highest and lowest values, so it is their average, not their difference.

Step 2. $\frac{14+4}{2} = \frac{18}{2}.$

Step 3. The midline is $y = \boxed{9}.$

Teaching note: together problems 2 and 3 recover A and D without knowing the equation. That is the whole idea of reading parameters off a graph.

4. Period and B for Curve G.

(a) Step 1. One period is the horizontal distance between two *successive* maxima — peak to peak, not peak to trough.

Step 2. The maxima sit at $t = 2$ and $t = 10$, so the period is $10 - 2 = \boxed{8}.$

(b) Step 3. B is fixed by the period through $\text{period} = \frac{2\pi}{B}$, so $B = \frac{2\pi}{8}.$

Step 4. $B = \boxed{\frac{\pi}{4}}$, which matches the $\sin(\pi t/4)$ in the equation of Curve G.

5. Building the Ferris wheel model.

(a) Step 1. The rider travels around a circle, so the height swings a full diameter from bottom to top; the amplitude is the radius.

Step 2. $\frac{40}{2} = \boxed{20}$ metres.

(b) Step 3. One full turn is one period, and a turn takes 12 minutes, so $B = \frac{2\pi}{12}.$

Step 4. $B = \boxed{\frac{\pi}{6}}.$

Completing the model: the lowest point is 3 metres up, so the midline is $3 + 20 = 23$, and boarding at the bottom means starting at the minimum — hence the leading minus sign in $h(t) = -20 \cos(\pi t/6) + 23.$

6. Height after four minutes.

Step 1. The time is given, so substitute — and keep the calculator in radian mode, because $\pi t/6$ is a radian measure.

Step 2. $\frac{\pi(4)}{6} = \frac{2\pi}{3}$, and $\cos \frac{2\pi}{3} = -\frac{1}{2}.$

Step 3. $h(4) = -20 \left(-\frac{1}{2}\right) + 23 = 10 + 23 = \boxed{33}$ metres.

Common wrong answer: about 3.05 — the cosine was taken in degrees, which returns a number near the bottom of the wheel and looks plausible. Radian mode is not optional here.

7. When the rider is level with the centre.

Step 1. $h(t) = 23$ makes the cosine term zero, so the question reduces to $\cos(\pi t/6) = 0$ — and cosine is zero at more than one place per cycle, so expect more than one answer.

Step 2. $\cos \theta = 0$ when $\theta = \frac{\pi}{2}$ or $\theta = \frac{3\pi}{2}$ within one revolution.

Step 3. Setting $\frac{\pi t}{6} = \frac{\pi}{2}$ gives $t = 3$; setting $\frac{\pi t}{6} = \frac{3\pi}{2}$ gives $t = 9$.

Step 4. Both lie in $0 \leq t < 12$, so the times are $\boxed{t = 3 \text{ and } t = 9}$ minutes — once on the way up, once on the way down.

Grading note: an answer of $t = 3$ alone is the error this problem is built to catch. A horizontal line through the midline cuts one cycle of the curve twice.

8. Building the harbour tide model.

(a) **Step 1.** High and low water are the maximum and minimum, so the amplitude is half their difference: $\frac{12 - 4}{2}$.

Step 2. $A = \boxed{4}$.

(b) **Step 3.** The midline is their average: $\frac{12 + 4}{2}$, so $D = \boxed{8}$.

(c) **Step 4.** High tide at $t = 3$ and the next low at $t = 9$ are *half* a period apart, so half a period is 6 hours and the full period is 12 hours.

Step 5. $B = \frac{2\pi}{12} = \boxed{\frac{\pi}{6}}$, giving $d(t) = 4 \cos\left(\frac{\pi(t - 3)}{6}\right) + 8$.

Common wrong answer: $B = \pi/3$, from treating the six hours between high and low as a whole period. Peak to trough is always half.

9. Parameters of a described sinusoid.

(a) **Step 1.** Same three formulas as Curve G, now with the maximum and minimum given in words instead of drawn.

Step 2. Amplitude = $\frac{18 - 2}{2} = \boxed{8}$.

(b) **Step 3.** Midline = $\frac{18 + 2}{2} = \boxed{10}$.

(c) **Step 4.** The maximum at $x = 1$ and the *next* minimum at $x = 7$ are half a period apart, so half the period is 6 and the period is $2(6) = \boxed{12}$.

Common wrong answer: 6 — the maximum-to-minimum gap read as a full period. Sketching one arch is the fastest way to see the factor of two.

10. Depth at $t = 5$.

Step 1. The model is built, so this is substitution: put $t = 5$ into $d(t) = 4 \cos\left(\frac{\pi(t - 3)}{6}\right) + 8$.

Step 2. $\frac{\pi(5 - 3)}{6} = \frac{\pi}{3}$, and $\cos \frac{\pi}{3} = \frac{1}{2}$.

Step 3. $d(5) = 4\left(\frac{1}{2}\right) + 8 = 2 + 8 = \boxed{10}$ feet.

11. All times when the depth is 10 feet.

Step 1. Subtract the midline first so the cosine stands alone: $4\cos\left(\frac{\pi(t-3)}{6}\right) = 2$, so $\cos\left(\frac{\pi(t-3)}{6}\right) = \frac{1}{2}$.

Step 2. Cosine equals $\frac{1}{2}$ at $\theta = \pm\frac{\pi}{3}$ within one revolution — two angles, because cosine is symmetric about $\theta = 0$.

Step 3. $\frac{\pi(t-3)}{6} = \frac{\pi}{3}$ gives $t-3 = 2$, so $t = 5$. $\frac{\pi(t-3)}{6} = -\frac{\pi}{3}$ gives $t-3 = -2$, so $t = 1$.

Step 4. Both lie in $0 \leq t < 12$, so the depth is 10 feet at $t = 1$ and $t = 5$ hours.

Step 5. How the graph predicts the count: $y = 10$ is a horizontal line above the midline but below the maximum of 12, so it cuts each single cycle of the curve exactly twice — once as the tide rises to high water at $t = 3$ and once as it falls again.

12. Synthesis: two models of one wheel.

(a) **Step 1.** Ana's model at $t = 5$: $\frac{\pi(5)}{6} = \frac{5\pi}{6}$ and $\cos \frac{5\pi}{6} = -\frac{\sqrt{3}}{2}$, so $h(5) = -20\left(-\frac{\sqrt{3}}{2}\right) + 23$.

Step 2. Ben's model at $t = 5$: $\frac{\pi(5-3)}{6} = \frac{\pi}{3}$ and $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$, so $h(5) = 20\left(\frac{\sqrt{3}}{2}\right) + 23$.

Step 3. Both give $10\sqrt{3} + 23$ metres, about 40.3 metres.

(b) **Step 4.** (*Open response — flagged for manual review; read the student's argument.*) A full-credit explanation contains all of this.

- Sine and cosine are the same wave a quarter period apart: $\sin\left(\theta - \frac{\pi}{2}\right) = -\cos\theta$. Ben's shift of 3 minutes is exactly a quarter of the 12-minute period, so his curve and Ana's are the same curve.
- A graph pins down the amplitude, the midline and the period uniquely — those are measurements of the picture and cannot be argued with.
- A graph does *not* pin down the horizontal shift C or the choice of sine against cosine, and it does not pin down the sign of A on its own: those three trade off against one another. Infinitely many (A, C) pairs describe one curve, because shifting by a whole period changes nothing and shifting by half a period is the same as flipping the sign of A .

Grading note: accept any argument that identifies amplitude, midline and period as determined and the phase/form as not. “They are both right because they both work” is the answer to push back on — ask *why* they both work.